

Vega Brakes Service Manual

Document: man-vega-brakes

Area: manuals

1 Brake System Overview

The Vega brake system provides hydraulic friction braking at all four wheels. The front circuit uses pads BP-7720 and rotors BR-7731, while the rear circuit uses pads BP-7722 and rotors BR-7733. Hydraulic actuation is provided by the master cylinder BMC-7780 and calipers BCAL-7760, with system fluid specified as BF-7790 and front flexible lines BH-7745. The comparable EV platform architecture is documented in the Marlin EV Brakes Service Manual .

2 Safety and Service Precautions

Brake dust may contain harmful materials, so never use compressed air to clean components and wear appropriate respiratory protection. Support the vehicle on rated stands before removing any wheel. Use only the specified brake fluid BF-7790 and never reuse fluid drained during service. Pump the pedal to seat the pads before moving the vehicle after any caliper or pad work.

3 Front Brake Sub-System

The front brake sub-system carries the majority of braking load using pads BP-7720, rotors BR-7731, and flexible hoses BH-7745. Inspect pad thickness, rotor runout, and hose condition during every brake service. Replace BP-7720 and BR-7731 as a set when either exceeds wear limits. See Marlin EV Brakes Service Manual :: Front Brakes for the comparable overview and Marlin EV Brakes Service Manual :: Front Pad and Rotor Specifications for limit values.

3.1 Front Brake Pad and Rotor Service

Use PROC-BRAKE-FR to service the front pads BP-7720 and rotors BR-7731 together when either component reaches its wear limit. Measure rotor thickness and lateral runout, and resurface or replace BR-7731 as required. Lubricate slide pins and apply approved anti-squeal compound to the BP-7720 backing plates. The equivalent combined service is in Marlin EV Brakes Service Manual :: Front Brake Pad and Rotor Service Procedure .

3.2 Front Brake Pad Replacement

Follow PROC-BRAKE-PAD to replace the front pads BP-7720 when rotors remain within specification. Retract the caliper piston carefully, install the new BP-7720 with hardware and shims, and verify free movement of the caliper on its slides. Pump the pedal to seat the pads before road test. The comparable procedure is in Marlin EV Brakes Service Manual :: Front Brake Pad Replacement Procedure .

3.3 Front Brake Hose Replacement

Use PROC-BRAKE-HOSE to replace the front flexible hose BH-7745 when it shows cracking, bulging, or fluid weeping. Cap the line to minimize fluid loss, install the new BH-7745 with new sealing washers, and route it clear of suspension travel. Bleed the affected circuit with fresh BF-7790 after installation. Inspect the hose at every service for chafing and age-related deterioration.

4 Rear Brake Sub-System

The rear brake sub-system uses pads BP-7722 and rotors BR-7733 to provide balanced braking and stability. Inspect for uneven wear that may indicate a sticking caliper or parking-brake adjustment fault. Replace BP-7722 and BR-7733 as a set when wear limits are reached. See Marlin EV Brakes Service Manual :: Rear Brakes and Marlin EV Brakes Service Manual :: Rear Pad and Rotor Specifications .

4.1 Rear Brake Pad and Rotor Service

Follow PROC-BRAKE-RR to service the rear pads BP-7722 and rotors BR-7733 together. On models with an integrated parking brake, use a scan tool or the manual procedure to retract the electric caliper piston before fitting new BP-7722. Measure and resurface or replace BR-7733 as required, then reset the parking brake. The equivalent procedure is in Marlin EV Brakes Service Manual :: Rear Brake Pad and Rotor Service Procedure .

5 Caliper Sub-System

The caliper sub-system clamps the pads against the rotors using the caliper assembly BCAL-7760. A seized piston or torn boot causes uneven pad wear, pulling, or drag. Inspect BCAL-7760 for fluid leaks and free slide-pin movement during service. See Marlin EV Brakes Service Manual :: Hydraulic Actuation for the comparable overview.

5.1 Brake Caliper Replacement

Use PROC-CALIPER to replace the caliper BCAL-7760 when the piston is seized or the body leaks. Disconnect the hydraulic line, mount the new BCAL-7760, and torque the bracket and banjo fittings to specification with new washers. Bleed the circuit thoroughly with BF-7790 and confirm a firm pedal. The comparable steps are in Marlin EV Brakes Service Manual :: Brake Caliper Replacement Procedure .

6 Hydraulic Sub-System

The hydraulic sub-system generates and distributes brake pressure through the master cylinder BMC-7780 using fluid BF-7790. Internal seal failure in BMC-7780 causes a slowly sinking pedal under steady pressure. Maintain BF-7790 at the specified moisture and concentration limits to protect hydraulic components. See Marlin EV Brakes Service Manual :: Hydraulic Lines and Fluid for the related overview.

6.1 Master Cylinder Service

Service the master cylinder BMC-7780 when the pedal sinks under sustained pressure or external leakage is found. Bench-bleed the replacement BMC-7780 before installation to remove trapped air. Torque the mounting nuts and line fittings to specification, then bleed the full system with fresh BF-7790. The comparable procedure is in Marlin EV Brakes Service Manual :: Brake Master Cylinder Service .

6.2 Brake Fluid Flush and Bleed

Use PROC-BRAKE-FLUID to flush and bleed the system with fresh BF-7790 at the specified interval or after any hydraulic repair. Bleed in the correct sequence, keeping the reservoir topped with BF-7790 to avoid drawing in air. Confirm a firm pedal and verify fluid moisture content is within limits before returning the vehicle to service.